Climate Change, Comparative Advantages, and Structural Transformation in Agriculture

Job Market Paper

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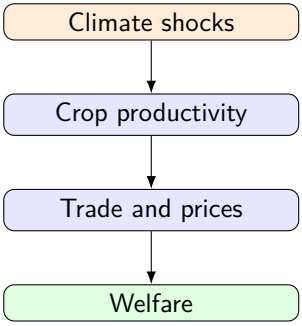
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Introduction

Why should we care?

- We study why climate change makes households poorer and how climate-induced productivity losses affect welfare when food expenditure shares vary with income.
- Climate change is not only an average productivity shock. It reshapes **where crops can be produced efficiently**.
- In agriculture, this means changes in comparative advantage, food prices, trade flows, and household welfare.



Key concern

Lower-income countries spend a larger share of income on food, so the same food-price shock can generate much larger welfare losses.

Research question

How do climate shocks affect welfare when food demand is non-homothetic?

- How do climate shocks reshape agricultural comparative advantages?
- How do productivity shocks transmit through production, trade, prices, and consumption?
- How much do income effects amplify welfare losses?
- How do adaptation and trade policies shape these losses?

Central mechanism

Climate shocks raise the relative price of food; because food expenditure shares are higher when income is lower, welfare losses are concentrated in lower-income economies.

Introduction

Antecedents and contribution

Existing literature

- Ricardian trade and agriculture: Eaton & Kortum (2002); Costinot et al. (2016).
- Climate change and trade adaptation: Gouel & Laborde (2021); Nath (2025).
- Structural change and non-homothetic demand: Comin et al. (2021).
- Food policy and climate shocks: Hsiao et al. (2026).

This paper

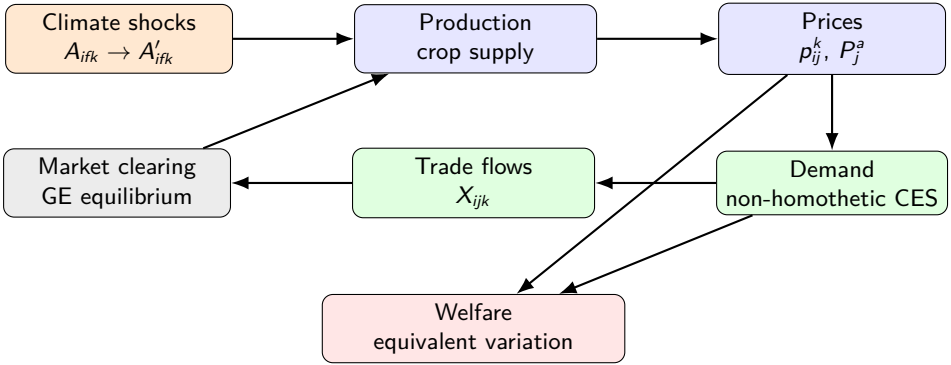
- Builds a multi-country Ricardian GE model with multiple crops.
- Embeds non-homothetic CES preferences.
- Measures welfare using money-metric utility and equivalent variation.
- Decomposes welfare changes into income, price, and structural change effects.

Contribution

Climate shocks reduce welfare mainly through higher food prices; losses are concentrated in lower-income economies, where food absorbs a large share of expenditure.

Theoretical Framework

Model structure: GE channels



Channel

Productivity shocks affect prices; prices and income jointly determine food expenditure shares; expenditure shares affect welfare.

Theoretical Framework

Countries, goods, agents, and markets

- Countries: importing country j and exporting country i.
- Goods:
  - agricultural crops indexed by $k \in \{1, \ldots, K\}$,
  - outside good 0 representing manufacturing and services.
- Agents:
  - A representative consumer in each country.
  - competitive producers/farmers.
  - landowners allocating parcels across crops.
- Endowments: labor L_i and land N_i.
- Land is divided into fields f and parcels $\omega \in [0, 1]$.

Market structure

Perfect competition; iceberg trade costs; labor mobile within country and immobile across countries.

Theoretical Framework

Assumptions

Demand side

- Non-homothetic CES preferences.
- Food expenditure shares vary with income.
- Representative household faces budget constraint.
- Agricultural goods are CES aggregates of crops and origins.

Supply side

- Outside good: constant returns to scale in labor.
- Crop production uses land and labor in fixed proportions.
- Parcel productivity follows a Fréchet distribution.
- Crop choice maximizes the value of marginal product of land.

Why these assumptions matter

They keep the model tractable while allowing climate shocks to affect comparative advantage, trade flows, prices, and welfare.

Stage 1: The representative household allocates expenditure between agriculture and the outside good:

$$\sum_{b\in\{0,a\}} (\beta_b)^{1/\alpha} \left(\frac{C_j}{L_j}\right)^{\varepsilon_b(1-\alpha)/\alpha} \left(\frac{C_j^b}{L_j}\right)^{(\alpha-1)/\alpha} = 1.$$

- $b \in \{0, a\}$: outside good and agricultural aggregate.
- β_b: preference weight.
- α: elasticity of substitution between broad goods.
- ε_b: non-homotheticity parameter ($\varepsilon_0 = 1$).

Engel channel

$$s_j^a = \beta_a \left(\frac{P_j^a}{P_j}\right)^{1-\alpha} \left(\frac{C_j}{L_j}\right)^{(1-\alpha)(\varepsilon_a-1)}.$$

When $\varepsilon_a < 1$, lower-income countries allocate a larger share of expenditure to food.

Stage 2: crop aggregator

$$C_j^a = \left[\sum_k (\beta_{jk})^{1/\kappa} (C_{jk})^{(\kappa-1)/\kappa} \right]^{\kappa/(\kappa-1)}.$$

Stage 3: varieties by exporter

$$C_{jk} = \left[\sum_i (\beta_{ijk})^{1/\sigma} (C_{ijk})^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}.$$

- C_{ijk}: consumption in country j of crop k from exporter i.
- κ: elasticity of substitution across crops.
- σ: elasticity of substitution across origins within a crop.

Demand for imported varieties

Demand depends on prices, crop substitution, origin substitution, and income per capita.

Crop production on parcel ω uses land productivity and labor requirements:

$$q_{ifk}(\omega) = \min\left\{A_{ifk}(\omega), \frac{L_{ifk}(\omega)}{\nu_{if}(\omega)}\right\}.$$

- $A_{ifk}(\omega)$: yield/productivity of parcel ω in field f.
- $\nu_{if}(\omega)$: labor requirement per unit of output.
- Production is Leontief: land productivity and labor are complements.

Crop choice

Land is allocated to the crop with the highest value of marginal product:

$$\pi_{ifk} = \Pr\left(p_{ik}A_{ifk}(\omega) > p_{i\ell}A_{if\ell}(\omega),\ \forall \ell \neq k\right).$$

Under the Fréchet distribution, crop supply is:

$$q_{ik} = \sum_{f} s_{if} A_{ifk} \left[\frac{(p_{ik} A_{ifk})^{\theta}}{(a_i)^{\theta} + \sum_{\ell} (p_{i\ell} A_{if\ell})^{\theta}} \right]^{(\theta-1)/\theta}.$$

- s_{if}: field area; a_i: country-specific labor-cost shifter, calibrated to match cultivated land.
- A_{ifk}: average productivity of crop k in field f.
- θ: dispersion of productivity across parcels.
- Smaller θ means larger productivity dispersion and stronger comparative advantage.

Climate shocks

Climate change enters only through field-crop productivity: $A_{ifk} \to A'_{ifk}$ (GAEZ potential yields under current and future climate).

Iceberg trade costs imply:

$$p_{ij}^k = \tau_{ij}^k p_i^k, \qquad \tau_{ij}^k \geq 1.$$

The value of exports of crop k from i to j is:

$$X_{ijk} = p_{ij}^k C_{ijk}.$$

Trade channel

Countries import from origins with lower delivered prices. Climate change changes delivered prices by changing productivity and domestic producer prices.

- Trade costs are not observed: they are absorbed into the variety weights β_{ijk}, calibrated to observed bilateral trade flows.
- Trade can mitigate climate losses by reallocating food supply toward less affected origins.

Factor market clearing

$$\sum_{f,k} L_{ifk}(\omega) = L_i, \qquad \sum_{f,k} N_{ifk}(\omega) = N_i.$$

Goods market clearing

$$q_i^k = \sum_j \tau_{ij}^k C_{ijk}, \qquad \forall i, k.$$

Equilibrium object

An equilibrium is a set of prices, production, consumption, and trade flows such that:

- consumers maximize utility,
- producers maximize profits,
- all goods and factor markets clear,
- trade flows are consistent with delivered prices and trade costs.

Theoretical Framework

Welfare

Welfare change is measured by the equivalent variation, at baseline prices P and post-shock utility u':

$$\Delta W = \frac{e(P, u') - e(P, u)}{e(P, u)}.$$

- $e(\cdot)$: expenditure function of the non-homothetic CES preferences.
- P, P': baseline and post-shock prices; u, u': baseline and post-shock utility.

Decomposition

$$1 + \Delta W = \underbrace{\frac{e(P', u')}{e(P, u)}}_{\text{income effect}} \times \underbrace{\frac{e(P, u)}{e(P', u)}}_{\text{price effect}} \times \underbrace{\frac{e(P, u')/e(P', u')}{e(P, u)/e(P', u)}}_{\text{structural change effect}}.$$

- Income effect: nominal income, including farm income.
- Price effect: cost of living at baseline utility.
- Structural change effect: income-driven shifts in expenditure shares; equals one under homothetic preferences.

Data and descriptive facts

Data

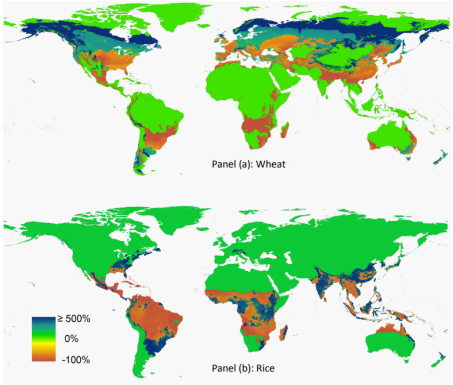
- **Sample:** 50 countries and 10 major crops.
- **Crops:** banana, soybean, cotton, sugarcane, maize, tomato, oil palm, wheat, rice, white potato.
- **Productivity:** GAEZ potential yields at 5-arcminute resolution (1.68 million fields), under the current climate and the climate projected for 2071–2100.
- **Climate scenarios:** Hadley CM3; SRES A1FI (benchmark), A2, B2, B1; high inputs, rain-fed, CO_2 fertilization.
- **Output, prices, and trade flows:** FAOSTAT.
- **Agricultural expenditure shares:** GTAP 9 Data Base (base year 2011).
- **GDP and income groups:** World Bank WDI.
- **Stylized facts:** World Input-Output Database (2000–2014).

Main limitation

Sample coverage is not exhaustive: included countries represent 89.7% of world crop output value, while selected crops account for 71.1%.

Key takeaways from facts to the model:

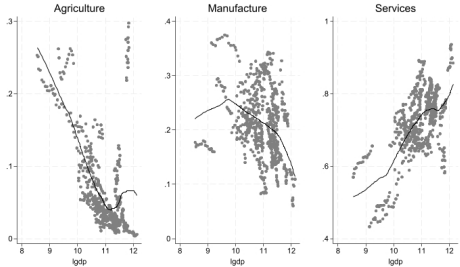
- Crop productivity varies substantially across space.
- This heterogeneity is the source of Ricardian comparative advantage.
- Climate change shifts these productivity patterns and therefore trade incentives.



Notes: Figure shows data from Costinot et al. (2016). Percent changes in yield due to climate change in GAEZ model for wheat and rice. Large green areas are those for which yields are zero both before and after climate change.

Key takeaways from facts to the model

- Agriculture's value-added share falls with income.
- Services become more important as countries develop.
- Lower-income countries remain more exposed to agricultural productivity and food-price shocks.



Notes: Each sub-figure displays the share of each sector in aggregate value added versus GDP per capita in 43 countries from 2000 to 2014 using World Input-Output Database.

Data and descriptive facts

Climate shocks and food expenditure shares

Income group	Yield change (%)	ΔA_i (% of GDP)	Agricultural share of consumption (%)
Low income	-34.0	-11.79	26.7
Lower-middle income	-34.2	-2.74	25.1
Upper-middle income	-33.0	-1.10	10.6
High income	-0.6	-0.01	3.1

Unweighted averages of country values within each income group. Hadley CM3, SRES A1FI.

Takeaway

Lower-income countries face the largest productivity losses **and** spend the largest share of their budget on food. The GDP-weighted world shock is -0.34% of world GDP.

Upper-level demand parameters are estimated by nonlinear least squares on the ratio of agricultural to non-agricultural expenditure shares, with $\varepsilon_0 = 1$:

$$\min_{\beta_a,\beta_0,\varepsilon_a} \sum_i \left[\frac{\beta_a}{\beta_0} \left(\frac{P_i^a}{P_i^0} \right)^{1-\alpha} \left(\frac{C_i}{L_i} \right)^{(1-\alpha)(\varepsilon_a - 1)} - \frac{s_i^a}{s_i^0} \right]^2, \qquad \beta_a + \beta_0 = 1.$$

- s_i^a / s_i^0: observed ratio of agricultural to non-agricultural expenditure shares (GTAP 9).
- P_i^a / P_i^0: relative price index.
- C_i / L_i: consumption per capita, measured as total expenditure divided by employment.
- α and ε_a are not separately identified: α is fixed at 0.5.

Empirical framework

Parameter values

Parameter	Description	Value
β_a	Preference weight, agricultural goods	0.62 (0.14)
β_0	Preference weight, non-agricultural goods	0.38 (0.14)
ε_a	Non-homotheticity, agricultural goods	0.083 (0.24)
α	Substitution, agricultural vs. other goods	0.50
ε_0	Non-homotheticity, non-agricultural goods	1.00
κ	Substitution between crops	2.82
σ	Substitution between origins of a crop	5.40
θ	Productivity heterogeneity within fields	2.46

Robust standard errors in parentheses.

Interpretation

$\varepsilon_a < 1$ is consistent with Engel's law: as real income increases, the share of expenditure spent on food declines. The implied income elasticity of food demand is about 0.55. κ, σ, and θ are taken from Costinot et al. (2016).

Why the counterfactual is informative?

- Climate shocks are introduced as productivity shocks: $A_{ifk} \rightarrow A'_{ifk}$.
- We compare the competitive equilibrium before and after the climate-induced productivity shocks, holding all other parameters fixed.
- Source of variation:
  - Spatial variation in potential crop yields across fields and countries.
  - Cross-country variation in income, prices, and expenditure shares.

Results and Discussion

Climate shocks and world welfare

Specification	World welfare change (% of income)	Demand calibration
Costinot et al. (2016)	−0.26	quasi-linear, ten crops only
This model, homothetic	−0.74	GTAP food shares
This model, non-homothetic	−0.83	food shares respond to income
Non-homothetic, employment-weighted	−3.77	

Main result

- Climate shocks reduce world welfare by 0.83% of income.
- Non-homothetic preferences amplify the losses, but most of the gap with Costinot et al. (2016) comes from food expenditure shares broader than the ten crops.
- Weighted by employment, the loss reaches 3.77%: populous lower-income economies lose the most.

Results and Discussion

Welfare losses by income group

Income group	Homothetic	Non-homothetic	Difference
Low income	−5.90	−6.23	−0.33
Lower-middle income	−7.51	−8.53	−1.02
Upper-middle income	−0.97	−1.09	−0.12
High income	−0.03	−0.04	−0.01
World	−0.74	−0.83	−0.09

Equivalent variation, % of income; GDP-weighted averages.

Takeaway

- Losses are concentrated in low- and lower-middle-income economies.
- Non-homothetic preferences raise the losses of every group, most in lower-middle-income countries.
- Largest losses: Sudan (−46.0%), Nigeria (−44.2%), Ghana (−42.6%). A few countries gain, e.g. Russia (+7.1%).

Results and Discussion

Mechanism: welfare decomposition

Income group	Agricultural price (%)	Price effect	Structural change effect	Income effect	Total ΔW
Low income	40.9	−7.87	−0.35	1.99	−6.23
Lower-middle	45.6	−8.10	−0.85	0.42	−8.53
Upper-middle	28.1	−1.05	−0.10	0.06	−1.09
High income	5.5	−0.09	0.00	0.05	−0.04
World	13.4	−0.83	−0.08	0.08	−0.83

Non-homothetic preferences, % of income; components sum to the total. Agricultural price: change in the consumer price index of agricultural goods.

- The **price effect** accounts for most of the losses in every income group.
- The structural change effect is smaller: about 10% of the world loss.
- Higher farm income offsets only part of the price effect, most in low-income countries.

Results and Discussion

Why losses are not proportional to productivity shocks

Country	ΔA_i (% GDP)	Price	Structural change	Income	Total ΔW
Sudan	−3.35	−36.28	−9.79	0.09	−45.98
Nigeria	−2.59	−36.88	−8.14	0.81	−44.21
Ghana	−6.37	−35.81	−8.46	1.69	−42.58
Malawi	−38.41	−26.17	−1.95	12.03	−16.09
Russia	−0.03	7.41	−0.23	−0.11	7.07

- **Sudan, Nigeria, Ghana:** consumer food prices more than double and food absorbs a large share of expenditure; structural change accounts for about a fifth of the loss.
- **Malawi:** largest productivity shock, but higher farm income offsets about half of the price effect.
- **Russia:** food prices fall and consumers gain.

Results and Discussion

Adaptation counterfactual

	Low	Lower-middle	Upper-middle	High	World
Welfare change (% of income)					
No adaptation	−6.23	−8.53	−1.09	−0.04	−0.83
Adaptation in lower-income countries	−1.76	−4.08	−1.04	−0.01	−0.51
Adaptation in all countries	−1.72	−3.82	0.38	0.17	−0.04
Agricultural price change (%)					
No adaptation	40.9	45.6	28.1	5.5	13.4
Adaptation in lower-income countries	14.0	19.6	26.9	4.0	10.4

Adaptation halves climate-induced yield losses (gains unchanged); costless yield improvements, an upper bound.

- Losses of lower-income countries fall by more than half, mainly because food prices rise much less.
- Adaptation elsewhere adds little for them: their losses stem mainly from their own productivity shocks.

Alternative SRES scenarios

Scenario	World NH	World Hom.	Low	Lower-middle
A1FI	−0.83	−0.74	−6.23	−8.53
A2	−0.77	−0.68	−4.70	−8.60
B2	−0.43	−0.38	−1.98	−4.66
B1	−0.20	−0.17	−2.39	−2.80

Alternative demand specifications

Specification	World	Low	Lower-middle
Benchmark	−0.83	−6.23	−8.53
Homothetic	−0.74	−5.90	−7.51
Lower substitutability	−1.00	−6.57	−9.29
Uniform food share	−1.46	−2.96	−5.24
Median imputation	−0.79	−5.54	−7.98

NH: non-homothetic; Hom.: homothetic. Equivalent variation, % of income; GDP-weighted averages.

- Losses increase with the severity of warming; lower-middle-income countries remain the largest losers.
- Giving every country the same food share halves the losses of low-income countries: cross-country differences in food shares shape the distribution of losses.

Results and Discussion

Trade policy counterfactuals

	30% export tax		No policy	Top-3 exporters	Full price insulation	
Income group	Maintained	Removed		50% tax	All crops	Staples
Low income	−6.07	−6.20	−6.23	−6.29	−6.37	−6.30
Lower-middle	−8.76	−8.39	−8.53	−8.86	−8.93	−8.53
Upper-middle	−1.17	−1.02	−1.09	−1.20	−1.28	−1.16
High income	−0.02	0.03	−0.04	−0.08	−0.07	−0.04
World	−0.85	−0.76	−0.83	−0.91	−0.93	−0.85

Equivalent variation under climate shocks, % of income. Price insulation: exporters set export taxes to absorb the climate-induced rise in their producer price. Staples: wheat, maize, rice.

- Removing export taxes raises welfare, but mostly independently of climate change.
- Export restrictions and price insulation amplify losses, particularly in lower-middle-income economies.

Summary and concluding remarks

1. Climate shocks reduce world welfare by 0.83% of income, with much larger losses in lower-income economies.
2. Higher agricultural prices account for most of the losses; the structural change effect is smaller, and higher farm income offsets only part of the price effect.
3. Non-homothetic preferences amplify the losses; cross-country differences in food expenditure shares shape their distribution.
4. Reducing climate-induced yield losses in lower-income countries substantially lowers their losses.
5. Removing export taxes raises welfare mostly independently of climate change; export restrictions and price insulation amplify the losses.

Policy message

Agricultural adaptation in vulnerable countries and avoiding trade restrictions that amplify the rise in food prices can help protect household welfare.

Q&A

Thank you!

Questions and Comments?